AUTOMATIC STAIRCASE LIGHT

A Mini Project Report submitted

to fulfill the design thinking learning Methodology

in

II-Year IV-Semester

By

J. Premalatha 24ME1A0407
J. Asha sri 24ME1A0424
K. Sandhya 24ME1A0486
D. Yamini 24ME1A04E8

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

RAMACHANDRA COLLEGE OF ENGINEERING (AUTONOMOUS)

(APPROVED BY AICTE - ACCREDITED BY NBA)

ELURU – 534007

(April - 2026)

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

RAMACHANDRA COLLEGE OF ENGINEERING (AUTONOMOUS)

(APPROVED BY AICTE - ACCREDITED BY NBA)

ELURU – 534007

(April - 2026)

CERTIFICATE

This is to certify that the project report entitled **"Automatic Staircase Light"** is being submitted by **J. Premalatha (24ME1A0407), J. Asha sri (24ME1A0424), K. Sandhya (24ME1A0486), D. Yamini (24ME1A04E8)** to fulfill the Mini project in Electronics and Communication Engineering to the Ramachandra College of Engineering, Eluru affiliated to JNTUK, Kakinada is a record of bonafide work carried out during the AY 2025-26.

Project Guide
B. Suneetha

Head of the Department
Dr. B Raghavaiah

Evaluator

ACKNOWLEDGEMENT

I would like to take this opportunity to express my deepest appreciation to the faculty coordinators for their valuable contributions and assistance with this design thinking learning

I would like to thank **B. Suneetha** for her guidance and support, especially for the valuable ideas and knowledge provided throughout the project, valuable comments and suggestions have been very useful in solving problems encountered during the project.

I wish to express our sincere thanks to **Dr. B. Raghavaiah**, Professor and Head of the Department of Electronics and Communication Engineering.

J. Premalatha (24ME1A0407)

J. Asha sri (24ME1A0424)

K. Sandhya (24ME1A0486)

D. Yamini (24ME1A04E8)

DECLARATION

I hereby declare that the work present in this project entitled "**AUTOMATIC STAIRCASE LIGHT**" submitted towards the fulfillment of the requirements for the mini project (design thinking learning Methodology) in Electronics and Communication in Ramachandra College of Engineering, Eluru is an authentic record of my work carried out for mini project, ECE Department, Ramachandra College of Engineering. The matter embodied in this report has not been submitted by us for the award of any other degree.

J. Premalatha (24ME1A0407)

J. Asha sri (24ME1A0424)

K. Sandhya (24ME1A0486)

D. Yamini (24ME1A04E8)

ABSTRACT

The Automatic Staircase Light using IR sensor is an energy-efficient and intelligent lighting system designed to automate the operation of lights in staircases. In many homes, offices, and public buildings, staircase lights are often left ON unnecessarily, leading to wastage of electrical energy. This project provides a solution by using infrared (IR) sensors and a microcontroller to detect human presence and control lighting automatically.

The system works by placing IR sensors at the bottom and/or top of the staircase. When a person approaches the stairs, the IR sensor detects the obstacle and sends a signal to the Arduino. Based on this signal, the Arduino activates the LEDs placed on each step. The LEDs glow in a sequential manner, creating a visually appealing and functional lighting system.

This project not only reduces energy consumption but also enhances safety, especially during nighttime. The use of sequential lighting makes the system modern and attractive. It is a simple, cost-effective, and efficient solution for smart home automation.

INTRODUCTION

Lighting systems play an important role in ensuring safety and convenience in daily life. Staircases, in particular, require proper lighting to avoid accidents. Traditionally, staircase lights are controlled manually using switches, which often leads to lights being left ON unnecessarily. This results in energy wastage and increased electricity bills.

To overcome this issue, automatic lighting systems have been introduced. These systems use sensors to detect human presence and automatically control the lighting. In this project, an IR sensor is used instead of a PIR sensor. IR sensors are widely used for obstacle detection and provide fast response when an object is detected in front of them.

The Automatic Staircase Light system uses an Arduino microcontroller to process signals from the IR sensor and control LEDs accordingly. The project demonstrates how embedded systems and sensors can be used to create smart and energy-efficient solutions. It is highly useful for educational purposes and real-world applications.

.

EXISTING SYSTEM

In the existing system, staircase lights are operated manually using switches. A two-way switch system is commonly used, where the user must turn ON the light before using the stairs and turn it OFF afterward. However, in many cases, users forget to switch OFF the lights, resulting in unnecessary power consumption.

Manual systems also lack convenience, especially in dark conditions where locating the switch can be difficult. This can lead to accidents, particularly for elderly people and children. Additionally, manual systems do not provide any intelligent control or automation.

Another limitation of the existing system is that it does not adapt to user movement. The lighting remains constant regardless of whether someone is present or not. Hence, there is a need for an automated system that can operate efficiently without human intervention.

PROPOSED SYSTEM

The proposed system introduces an automatic staircase lighting system using IR sensors and Arduino. The IR sensors are placed at strategic positions (bottom and/or top of the staircase) to detect the presence of a person. When a person is detected, the system automatically turns ON the lights.

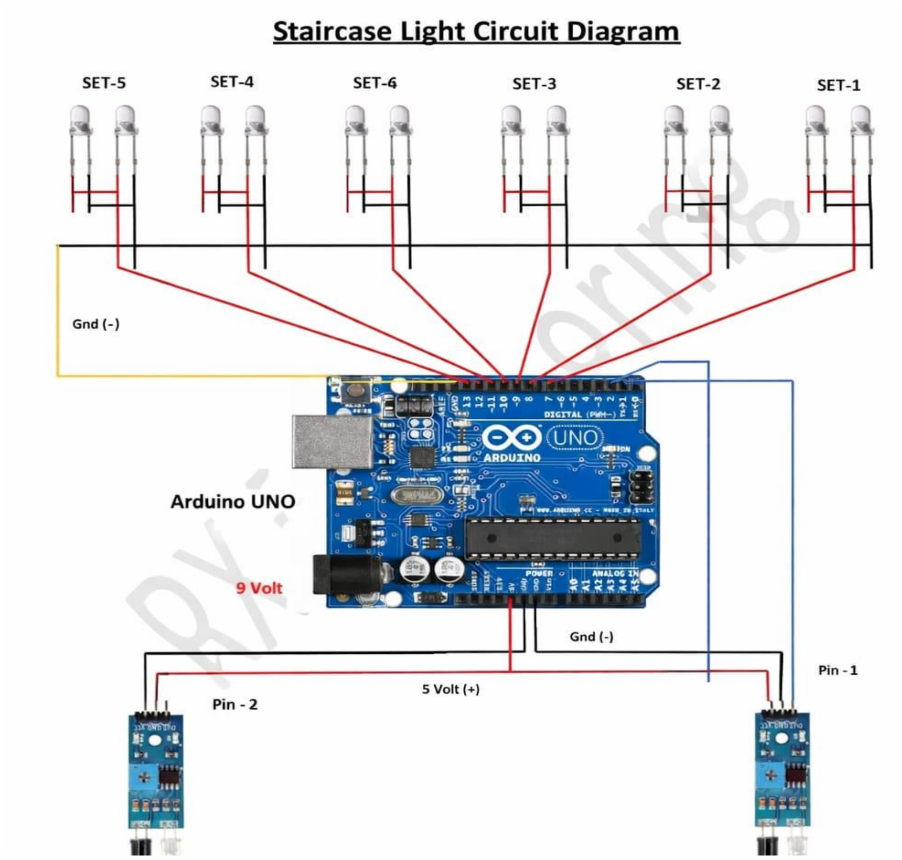
The LEDs are arranged on each step and glow sequentially when triggered. This creates a dynamic lighting effect and ensures better visibility while climbing or descending the stairs. When the person leaves the staircase, the lights turn OFF automatically after a delay or in reverse sequence.

This system eliminates the need for manual switches and reduces energy wastage. It also improves safety and provides a modern look to the staircase. The use of IR sensors ensures quick detection and reliable operation, making the system efficient and practical.

COMPONENTS:

Component	Quantity	Description
Arduino Uno	1	Microcontroller used to control the entire system
IR Sensor Module	2	Detects object/person using infrared rays
LEDs	10	Used as staircase lights for each step
Resistors (220Ω)	10	Protect LEDs from excess current
Breadboard	1	Used for temporary circuit connections
Jumper Wires and Connecting Wires	As required	Used to connect components together. Used for internal wiring in model
Foamboard	As required	Used to build staircase model
Glue	As required	For fixing staircase structure

CIRCUIT DIAGRAM AND EXPLANATION:



The IR sensor detects the presence of a person using infrared rays. When an object comes in front of the sensor, the reflected IR rays are detected, and the sensor outputs a signal (LOW or HIGH depending on module).

This signal is sent to the Arduino, which processes it and activates the LEDs. Each LED represents a step in the staircase. The Arduino turns ON the LEDs sequentially to simulate stair lighting. When no object is detected, the Arduino turns OFF the LEDs after a delay.

CODE:

```
int IRSensor1 = 1;
int IRSensor2 = 2;

void setup()
{
  pinMode (IRSensor1, INPUT);
  pinMode (IRSensor2, INPUT);
  pinMode (6, OUTPUT);
  pinMode (7, OUTPUT);
  pinMode (8, OUTPUT);
  pinMode (9, OUTPUT);
  pinMode (10, OUTPUT);
}
void loop()
{
  int statusSensor1 = digitalRead (IRSensor1);
  int statusSensor2 = digitalRead (IRSensor2);
  if (statusSensor1 == 0)
  {
     digitalWrite(6, HIGH);
    delay(1000);
     digitalWrite(7, HIGH);
    delay(1000);
     digitalWrite(8, HIGH);
    delay(1000);
     digitalWrite(9, HIGH);
```

```
  delay(1000);
   digitalWrite(10, HIGH);
  delay(1000);
   digitalWrite(6, LOW);
  delay(1000);
   digitalWrite(7, LOW);
  delay(1000);
   digitalWrite(8, LOW);
  delay(1000);
   digitalWrite(9, LOW);
  delay(1000);
   digitalWrite(10, LOW);
  delay(1000);
}
else
{
   digitalWrite(10, LOW);
   digitalWrite(9, LOW);
   digitalWrite(8, LOW);
   digitalWrite(7, LOW);
   digitalWrite(6, LOW);
}

if (statusSensor2 == 0)
{
   digitalWrite(10, HIGH);
  delay(1000);
   digitalWrite(9, HIGH);
```

```
   delay(1000);
    digitalWrite(8, HIGH);
   delay(1000);
    digitalWrite(7, HIGH);
   delay(1000);
    digitalWrite(6, HIGH);
   delay(1000);
    digitalWrite(10, LOW);
   delay(1000);
    digitalWrite(9, LOW);
   delay(1000);
    digitalWrite(8, LOW);
   delay(1000);
    digitalWrite(7, LOW);
   delay(1000);
    digitalWrite(6, LOW);
   delay(1000);
  }

  else
  {
    digitalWrite(10, LOW);
    digitalWrite(9, LOW);
    digitalWrite(8, LOW);
    digitalWrite(7, LOW);
    digitalWrite(6, LOW);
  }
}
```

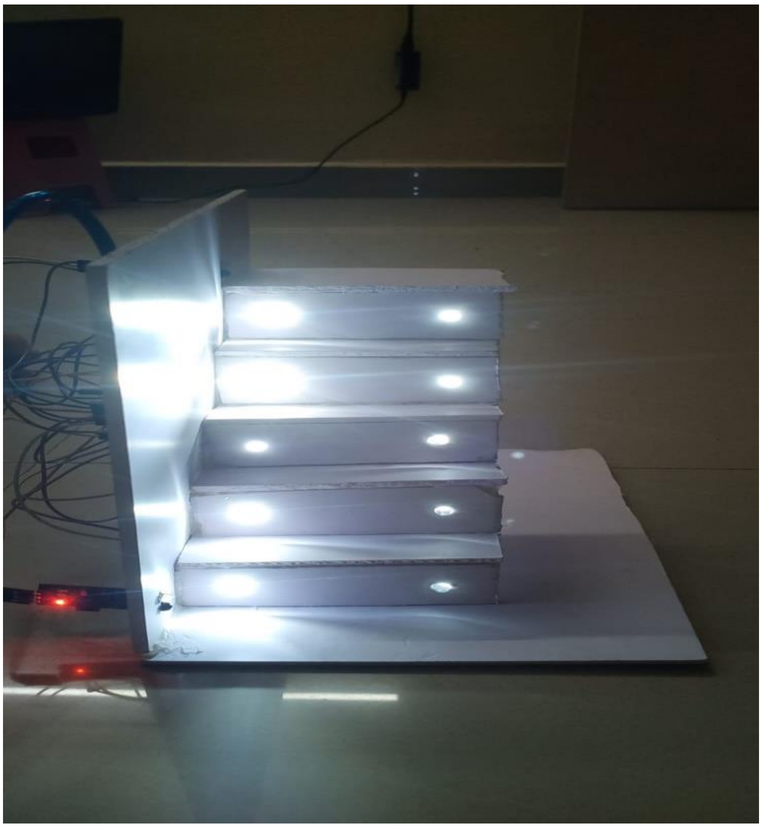
WORKING AND OUTPUT:

The working of the Automatic Staircase Light system is based on the detection of objects using an IR sensor and the control of LEDs using an Arduino microcontroller. The IR sensor continuously emits infrared rays and monitors their reflection. When no object is present, the emitted rays do not reflect back to the receiver, and the sensor remains in its normal state. However, when a person approaches the staircase, the infrared rays reflect back from the person's body, and the sensor detects the presence of the object. This detection generates a signal which is sent to the Arduino for further processing.

Upon receiving the signal from the IR sensor, the Arduino activates the LEDs placed on each step of the staircase. Instead of turning all lights ON at once, the system is programmed to glow the LEDs sequentially from the bottom step to the top step. This creates a smooth lighting effect that resembles real staircase illumination. The sequential lighting not only improves visibility but also enhances the visual appearance of the model. After all the LEDs are turned ON, they remain in that state for a predefined time delay, allowing the person to safely move across the staircase.

Once the person leaves the staircase or after the delay time is completed, the Arduino automatically turns OFF the LEDs in reverse order, i.e., from the top step to the bottom step. This gradual turning OFF of lights gives a dynamic effect and ensures that the lights are not left ON unnecessarily. The output of this system is an automatic, energy-efficient staircase lighting setup where the lights operate only when required. The project successfully demonstrates how sensors and microcontrollers can be used to develop smart systems that improve safety, convenience, and energy conservation.

TESTING THE SYSTEM:



APPLICATIONS:

1. The system is widely used in **residential buildings and individual homes** to automate staircase lighting. It ensures that lights turn ON only when a person is present and turn OFF automatically after use, thereby reducing unnecessary electricity consumption and improving energy efficiency.
2. It is especially useful in **houses where lights are often left ON accidentally**, helping to avoid power wastage and lowering electricity bills without requiring any manual effort from the user.
3. The project can be effectively implemented in **commercial buildings, offices, and workplaces**, where automatic lighting systems improve convenience for employees and eliminate the need for manual switching in frequently used staircases and corridors.
4. It is also suitable for **public places such as shopping malls, hospitals, and railway stations**, where there is continuous movement of people. In such environments, automatic lighting ensures proper illumination at all times while optimizing energy usage.
5. The system contributes significantly to **energy conservation**, as it operates only when required. This makes it an important part of modern energy-saving technologies used in smart infrastructure and sustainable building designs.
6. It enhances **safety and visibility**, especially in dark conditions or during nighttime, by automatically lighting up staircases when movement is detected, thereby reducing the chances of slips, falls, and accidents.
7. The project is highly beneficial for **elderly people and children**, as it removes the need to locate and operate switches, providing immediate lighting and making staircases safer and easier to use.

CONCLUSION:

The Automatic Staircase Light using IR sensor is a simple yet effective solution for energy-efficient lighting. It eliminates the need for manual operation and ensures that lights are used only when required. This helps in reducing electricity consumption and promoting sustainable energy usage.

The project demonstrates the practical application of sensors and microcontrollers in real-world problems. It is easy to implement, cost-effective, and highly reliable. The use of sequential lighting adds an aesthetic appeal, making it suitable for modern homes and buildings.

In conclusion, this project successfully achieves automation, safety, and energy efficiency. It serves as an excellent example of how technology can be used to improve everyday life.